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INVITED PAPER

Quantum differential cryptanalysis

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Abstract
The work is devoted to the study quantum versions of the differential cryptanalysis based on using a combination of the quantum minimum/maximum search algorithm and the quantum counting algorithm. We have estimated the complexity and the required resources for applying the quantum differential and quantum linear cryptanalysis to searching round keys of block ciphers. It is shown that the implementation of the quantum linear method requires less logical qubits than for the implementation of the quantum differential method. The acceleration of calculations due to "quantum parallelism" in the quantum differential cryptanalysis, based on a combination of Grover's quantum algorithms and quantum counting algorithm, is apparently absent, because the using of quantum counting as "subprogram" in the Grover algorithm eliminates quantum acceleration, as far as $O(\sqrt{K}) \cdot O(\sqrt{K}) \approx O(K)$.

Keywords Symmetric cryptography · Quantum attacks · Differential cryptanalysis · Block ciphers · Grover's algorithm · Quantum counting

1 Introduction

Quantum computing is one of the areas of quantum technology that has been developing since the end of the 20th century. Attempts to accelerate calculations by using quantum parallelism are used in cryptanalysis, including quantum differential and quantum linear cryptanalysis.

It was shown in [3–5] that using the Bernstein-Vazirani and Simon's quantum algorithms, we could search for quasi-linear structures of Boolean functions and difference relations with significant characteristics with polynomial complexity.

The works [6,7] are devoted to quantum differential cryptanalysis based on a combination of Grover's algorithm [8] and quantum counting algorithm [9,10]. Unfortunately, there are some inaccuracies and gaps.

In this regard, the main goal of this work is to describe in detail the quantum differential cryptanalisis methods, to obtain estimates of subkeys searching complexity, and also to display the main problems that may arise during the application of these methods, provided that an ideal universal quantum computer comes into the world and when the difference relationships, with corresponding characteristics, are already known.

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2 Quantum differential cryptanalysis

The differential cryptanalysis was introduced in [1]. In this section there is a method for recovering the first round key of a block cipher $E : V_n \times V_m \to V_m$ with using quantum algorithms is presented by following the [7].

We denote a composition of R round transformations $E^1 : V_k \times V_m \to V_m$ by $E(key, P) = C$, i.e. $E \equiv (E^1)^R = E^R$. We want to find one of the $K = 2^n$ binary vectors of length n, the secret key denoted by key.

Let's denote $(a, b)_r$, $a, b \in V_m$, $r \in \overline{1, R}$ – the difference relations for E^r. The difference relation $(a, b)_r$ on some fixed key is performed with probability $p_{(a,b)_r}$, i.e.

$$P(E^r(key, P) \oplus E^r(key, P \oplus a) = b) = p_{(a,b)_r}, \qquad (1)$$

and moreover

$$p_{(a,b)_r} = \max_{x,y \in V_m \setminus \{0\}} p_{(x,y)_r}.$$

The subkey of the r-th round is denoted by $K_r \in V_k$.

Suppose that we have difference $(a, b)_{R-1}$ with the difference characteristic $p_{(a,b)_{R-1}}$ and all $N = 2^m$ possible pair

blocks of plain text and cipher text (P_i, C_i), $P_i, C_i \in V_m$, $i \in \overline{0, 2^m - 1}$ received on the same secret key.

Since the difference characteristic is $p_{(a,b)_{R-1}} = \max_{x,y \in V_m \setminus \{0\}} p_{(x,y)_{R-1}}$, it is required to find such a candidate for the first round key K_1 that the probability of the difference relation "If $P'_i \oplus P'_j = a$, then $C_i \oplus C_j = b$" on $R - 1$ rounds will be the maximum, $P'_i = E^1(K_1, P_i)$ is the result of applying one iteration of the encryption algorithm to the known block P_i on the round key K_1.

Let $Z(K_1)$ is the number of pairs of plaintext blocks (P'_i, P'_j), $i, j \in \overline{0, 2^m - 1}$ on which the property "If $P'_i \oplus P'_j = a$, then $C_i \oplus C_j = b$" is performed,

$$Z(K_1) = \sum_{i,j=0}^{2^m-1} Ind(C_i \oplus C_j = b | P'_i \oplus P'_j = a),$$

or, if we denote remaining $R - 1$ rounds on the true secret key $C_i = E^{R-1}(key, P'_i)$, then

$$Z(K_1) = \sum_{i=0}^{2^m-1} Ind(E^{R-1}(key, P'_i) \oplus E^{R-1}(key, P'_i \oplus a) = b),$$

where $Ind(x) \in \{0, 1\}$ is an indicator that the logical expression x is performed.

We assume that for wrong candidates of the first round key the probability of the difference has a uniform distribution:

$$P(\text{"If } P'_i \oplus P'_j = a, \text{ then } C_i \oplus C_j = b\text{"}) \sim Be(1/2^{m-1}).$$

If on the round key K_1 the probability of the difference (a, b) implementation is maximum, then on the round key K_1 the value of $Z(K_1)$ is also maximum.

Thus the task of finding probable candidates for K_1 is equivalent to optimizing the value of $Z(K_1)$, i.e. finding such round key K_1 on which $Z(K_1) \to max$.

The quantum differential method of cryptanalysis is based on a combination of the quantum algorithm for finding the minimum [11] (in our case finding the maximum) and the quantum counting algorithm (see [9,10]), due to which computation acceleration is expected compared to the classical case.

In the classical case, in order to find $Z(K_1)$ it is necessary to check all known possible pairs (P_i, C_i), $P_i, C_i \in V_m$, $i \in \overline{0, 2^m - 1}$, i.e. for each pair of blocks (P_i, C_i), calculate $P'_i = E^1(K_1, P_i)$ and check if the property "If $P'_i \oplus P'_j = a$, then $C_i \oplus C_j = b$" is performed.

In the quantum case, we want to use quantum parallelism, i.e. get a superposition of all possible iterative keys K_1 and the corresponding values $Z(K_1)$. Initial quantum state - state

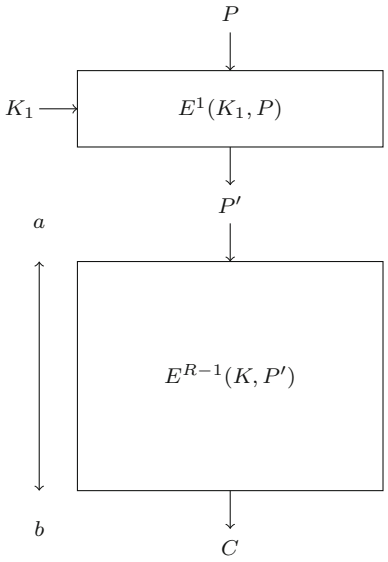


Fig. 1 Scheme for the K_1 searching

of uniform distributed superposition

$$\sum_{K_1} \frac{1}{\sqrt{2^k}} |K_1\rangle |Z(K_1)\rangle,$$

to which we will apply the amplitude amplification procedure to search for candidates for the K_1.

First of all, from the known $N = 2^m$ pairs of blocks (P_i, C_i), $i \in \overline{0, N - 1}$, it is necessary to prepare the state

$$\sum_{i=0}^{2^m-1} \frac{1}{\sqrt{2^m}} |P_i\rangle |C_i\rangle.$$

This state can be obtained by applying generalized $CNOT(C|t)$ gates, in which qubit t is controlled by the set of qubits C. Generalized gates $CNOT(C|t)$ can be implemented without using ancilla qubits (see [12]), therefore, we will consider generalized gates $CNOT(C|t)$ as one logic gate.

As an example, let's consider the case when blocks of plaintext and ciphertext are three-bit vectors defined in the Table 1.

Quantum circuit for preparing $\sum_{i=0}^{7} \frac{1}{\sqrt{2^3}} |P_i\rangle |C_i\rangle$ represented in Fig. 2.

The Fig. 2 uses the generalized elements of $CNOT(C|t)$ with several controlled qubits (see [12], section 4.3), which means the composition of several $CNOT(C|t)$ by analogy with the diagram in the Fig. 3.

Let's consider a way to prepare the state $|K_1\rangle |Z(K_1)\rangle$ for an arbitrary first round key $K_1 \in V_k$.

In [7], the quantum counting procedure is used for estimating $Z(K_1)$, i.e. for estimating weight of Boolean function

Table 1 The table of values (P_i, C_i) for an example of constructing a quantum circuit that prepares the quantum state $\sum_{i=0}^{7} \frac{1}{\sqrt{2^3}}|P_i\rangle|C_i\rangle$

The pair number i	P_i	C_i
0	000	001
1	001	010
2	010	011
3	011	100
4	100	101
5	101	110
6	110	111
7	111	000

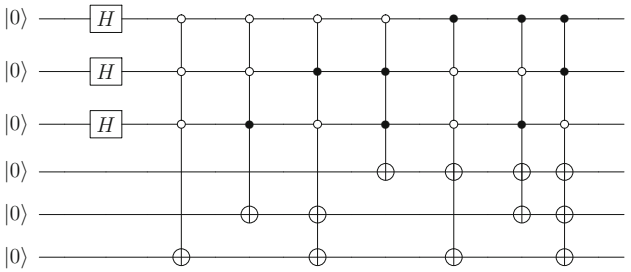


Fig. 2 Quantum circuit for preparing $\sum_{i=0}^{7} \frac{1}{\sqrt{2^3}}|P_i\rangle|C_i\rangle$. The upper three qubits correspond to the block P_i, the bottom three qubits correspond to the block C_i, specified in Table 1. The highest bit orders of P_i and C_i are on the top, the lower orders are on the bottom. There are no operations required to initialize $|111\rangle|000\rangle$ since the bottom three qubits are already initialized in the states $|0\rangle$

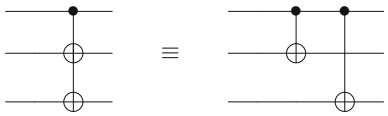


Fig. 3 Element $CNOT$ with several controlled qubits

f_{K_1}, that check whether the difference relation $(a, b)_{R-1}$ on blocks $(E^1(K_1, P_i), C_i)$, $i \in \overline{0, 2^m - 1}$ and round key K_1 has performed.

For fixed value of K_1 $in V_k$ we have $f_{K_1} : V_m \to V_1$. In the general case, when $K_1 \in V_k$ is unknown, $f_{K_1} : V_k \times V_m \to V_1$.

Note that [7] does not contain a detailed description of the implementation of the function f_{K_1} (see [7], section 3.2). There are understated estimates of the complexity and necessary quantum resources (qubits and gates).

For preparing the quantum state $\sum_{i=0}^{N-1} \frac{1}{\sqrt{N}}|P_i\rangle|C_i\rangle$, $O(N)$ quantum operations are required. To implement f_{K_1}, we need to prepare $\sum_{i=0}^{N-1} \frac{1}{\sqrt{N}}|P_i\rangle|C_i\rangle$ twice. Figure 4 shows a quantum circuit that implements f_{K_1}.

In order to get the state $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|Z(K_1)\rangle$ it is enough to perform the quantum counting procedure as in the Fig. 5 (except qubit measurement operations) and estimate the weight of Boolean function f_{K_1} (Fig. 4). Thus we have obtained a superposition of iterative keys and some estimates θ_{K_1} (see quantum counting, [9,10,12]):

$$\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|\theta_{K_1}\rangle.$$

In order to get $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|Z(K_1)\rangle$ from $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|\theta_{K_1}\rangle$ we have to evaluate $N \sin^2(\theta_{K_1}/2)$, but there is enough to get only $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|\theta_{K_1}\rangle$. Instead of comparing $Z(K_1)$ and $Z(K_1')$, we will compare θ_{K_1} and $\theta_{K_1'}$

Since in Grover's algorithm (see [8,12]) when we search one of M correct solutions among the N possibles $\sqrt{\frac{M}{N}} = \sin\left(\frac{\theta}{2}\right)$, $\theta \in [0, 1)$, if $\theta_1 > \theta_2$, then $N \sin^2\left(\frac{\theta_1}{2}\right) > N \sin^2\left(\frac{\theta_2}{2}\right)$.

Remark 1 In order not break the reversibility of quantum computations, if the procedure for measuring qubits in the Fig. 5 would be omitted, then instead of $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|\theta_{K_1}\rangle$ we get $\sum_{K_1} \sum_{theta_{K_1}} \frac{1}{\sqrt{2^k}}\sqrt{P(\theta_{K_1})}|K_1\rangle|\theta_{K_1}\rangle$, where $P(\theta_{K_1})$ is the probability that when measuring qubits in the Fig. 5 will get the value θ_{K_1}, one of the 2^{w_N} possible.

For simplicity, we will not take this fact into account, we will assume that if we omit the measuring qubits on the quantum circuit Fig. 5, we will still get $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|\theta_{K_1}\rangle$. This assumption will not affect the final conclusions.

The complexity of the quantum counting procedure is $O(2^{w_N})$ Grover iterations with respect to the Boolean function f_{K_1}. The probability of success of the quantum counting procedure is at least $4/\pi^2$.

When we use the Grover algorithm (see [8,12]) for searching one of the M correct solutions among N possible, $\sqrt{\frac{M}{N}} = \sin\left(\frac{\theta}{2}\right)$, in order to obtain the number $Z(K_1)$ from the estimates θ_{K_1} after applying the quantum counting procedure, it is necessary to calculate $F(\theta) = N \sin^2\left(\frac{\theta}{2}\right)$. However, to search for candidates for the first iteration key K_1, it is not necessary to calculate $F(\theta) = N \sin^2\left(\frac{\theta}{2}\right)$, since $\theta \in [0, 1)$ and if $\theta_1 > \theta_2$, then $N \sin^2\left(\frac{\theta_1}{2}\right) > N \sin^2\left(\frac{\theta_2}{2}\right)$.

To searching for candidates for the first round key K_1, it is enough to get the state $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|\theta_{K_1}\rangle$. Next, the amplitude amplification algorithm (Grover's algorithm) is used for searching round keys with a maximum value of $Z(K_1)$.

Quantum Differential Cryptanalysis

Input. Block cipher with round encryption $E : V_k \times V_m \to V_m$, $N = 2^m$ pairs of plain text and cipher text blocks (P_i, C_i), $P_i, C_i \in V_m$, $i \in \overline{0, 2^m - 1}$ received on the same secret key, difference relation $(a, b)_{R-1}$ with $p_{(a,b)_{R-1}}$, such that two simple hypotheses can be distin-

Table 2 The table of values of the Boolean function f_{K_1}, by which we estimate the value of $Z(K_1)$. Here we denote $(x_1, x_2, \ldots, x_m)$ - binary representation of the number $i \in \overline{0, 2^m - 1}$, and $f(x_1, x_2, \ldots, x_m)$ is the column of values of the required Boolean function. $P'_i = E^1(K_1, P_i)$ - the result of applying one iteration of the encryption algorithm to the known plaintext block P_i on the round key K_1. The table is useful for building quantum circuit that implements the Boolean function f_{K_1}

Number of a pair P_i, C_i	$i_{binary} = (x_1, x_2, \ldots, x_m)$	$f(x_1, x_2, \ldots, x_m)$
$i = 0$	$00 \ldots 00$	$Ind(E^{R-1}(key, P'_0) \oplus E^{R-1}(key, P'_0 \oplus a) = b)$
$i = 1$	$00 \ldots 01$	$Ind(E^{R-1}(key, P'_1) \oplus E^{R-1}(key, P'_1 \oplus a) = b)$
$i = 2$	$00 \ldots 10$	$Ind(E^{R-1}(key, P'_2) \oplus E^{R-1}(key, P'_2 \oplus a) = b)$
$i = 3$	$00 \ldots 11$	$Ind(E^{R-1}(key, P'_3) \oplus E^{R-1}(key, P'_3 \oplus a) = b)$
$\vdots$	$\vdots$	$\vdots$
$i = N - 1$	$11 \ldots 11$	$Ind(E^{R-1}(key, P'_{N-1}) \oplus E^{R-1}(key, P'_{N-1} \oplus a) = b)$

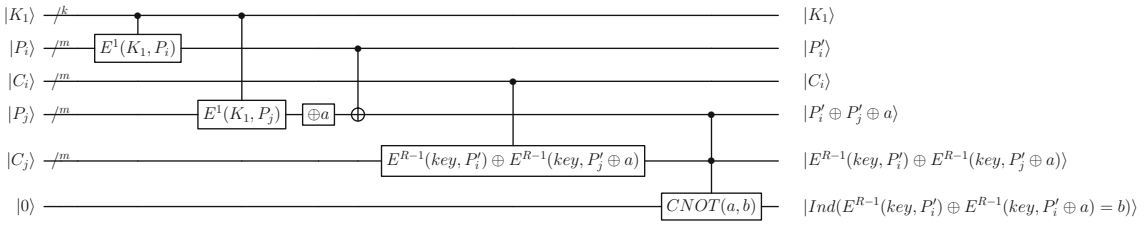


Fig. 4 Implementation of f_{K_1} as a quantum circuit. $E^1(K_1, P_i)$ - one iteration of the encryption function on the key K_1 and the plaintext block P_i, which is theoretically possible without the using of ancilla qubits (see [13,14]) The operation $\oplus a$ is the application of one-qubit gates X on those qubits whose numbers correspond to "1" bits of the difference $a \in V_m$. The operation $E^{R-1}(key, P'_i) \oplus E^{R-1}(key, P'_i \oplus a)$ is performed using m standard two-qubit operations $CNOT$, and $CNOT(a, b)$ - generalized $CNOT$, inverting the lower qubit, in which the differences a and b are the control vectors

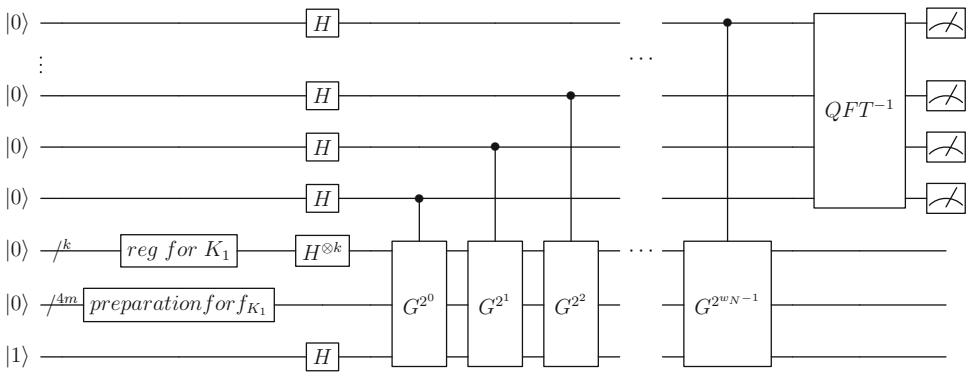


Fig. 5 Quantum circuit for preparation the state $\sum_{K_1} \frac{1}{\sqrt{2^k}} |K_1\rangle |\theta_{K_1}\rangle$ (see quantum counting, [9,10,12]). For simplicity we define that the control register contains $w_N = \lceil m/2 \rceil + 3$ qubits (we could improve the accuracy of the quantum counting by using additional control qubits)

guished by classical statistical methods

H_0:P("If $P'_i \oplus P'_j$=a, then $C_i \oplus C_j$=b") $\sim Be(1/2^{m-1})$;
H_1:P("If $P'_i \oplus P'_j$=a, then $C_i \oplus C_j$=b") $\sim Be(p_{(a,b)_{R-1}})$,

provided $1/2^{m-1} < p_{(a,b)_{R-1}}$, with acceptable error probabilities. The case when $p_{(a,b)_r} = \min_{x,y \in V_m \setminus \{0\}} p_{(x,y)_r}$ and $p_{(a,b)_{R-1}} = 0$ is known as the impossible differential method.
Output. Round key K_1 with a maximum value of $Z(K_1)$.

Preparation step

1: Prepare two registers with quantum states $\sum_{i=0}^{N-1} \frac{1}{\sqrt{N}} |P_i\rangle |C_i\rangle$. The complexity of this stage of $O(N)$ quantum operations.
2: Select a random round key K, calculate the value of $Z(K)$. At this step, we can get the exact value of $Z(K)$ by analysing all N pairs of blocks of open and encrypted text on a classic computer, then calculate the initial value $\theta_{current} := \theta_K = 2 \arcsin \sqrt{\frac{Z(K)}{N}}$.
This step can be performed with the complexity of $O(2^{\lceil \frac{m}{2} \rceil + 3})$ Grover iterations relatively to the Boolean

function f_K, with a probability of at least $4/\pi^2$ estimate $\widetilde{\theta}_{current}$, which has a predicted error level $\Delta\widetilde{\theta}_{current} \leq 2^{-3}$ (see [9,10,12]).

Search procedure

1: Initialize k qubits in state $|0\rangle$, apply the $H^{\otimes k}$ operator to these qubits, consisting of k standard Hadamard gates H. Thus, we obtain an uniform distributed superposition of keys $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle$.
2: Execute the quantum counting as in the Fig. 5 with using $w_N = \lceil m/2 \rceil + 3$ control qubits to determine $||f_{K_1}||$ without applying measurements of qubits.

$$f_{K_1} : V_k \times V_m \to V_1,$$
$$f_{K_1}(P_i) = Ind(E^{R-1}(key, P_i') \oplus E^{R-1}(key, P_i' \oplus a) = b).$$

A quantum circuit implementing f_{K_1} is shown in the Fig. 4.
The complexity of the quantum counting is $O(2^{w_N})$ Grover iterations relatively to the Boolean function f_K, with a probability of at least $4/\pi^2$ we get a superposition of the estimates $\widetilde{\theta}_{K_1}$ which has a predicted error level $\Delta\widetilde{\theta}_{K_1} \leq 2^{-3}$.
Thus, after quantum counting we obtain a superposition of round keys and the corresponding estimates $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|\widetilde{\theta}_{K_1}\rangle$ (other registers are omitted for clarity, because they are not involved in the further search procedure), at this step we need $k + 4m + 1 + w_N$ logical qubits.
3: The value $\theta_{current}$ is used as a threshold value, the set of all possible round keys $K \in V_k$ is divided into two classes: the first class of "bad" keys $\theta_K \leq \theta_{current}$ and the second class of "good" keys $\theta_K > \theta_{current}$. This classification can be described by the Boolean function

$$g(\widetilde{\theta}_{K_1}, \theta_{current}) = Ind(\widetilde{\theta}_{K_1} > \theta_{current}),$$

$$g : V_{w_N} \times V_{w_N} \to V_1.$$
The Boolean function $g(\widetilde{\theta}_{K_1}, \theta_{current})$ could be implemented as follows. We could initialize one ancilla qubit $|q\rangle = |1\rangle$, and consider it as high order bit of binary decomposition $\theta_{current}$. If after subtraction in the binary number system $\theta_{current} - \widetilde{\theta}_{K_1}$, the high order bit is changed and will be "zero" (we get $|q\rangle = |0\rangle$), then $\widetilde{\theta}_{K_1} > \theta_{current}$. This method of comparing two integers was proposed by [18] and could be implemented by inverting a quantum circuit that implements the operation of adding two integers (see [19–21]).
Perform the quantum counting procedure as in the Fig. 5 with using $w_K = \lceil k/2 \rceil + 3$ control qubits – with a probability of at least $4/\pi^2$ and complexity $O(2^{w_K})$ Grover iterations relatively to the Boolean function g, we obtain the estimate θ_g, by which we find

$$||g|| \approx \widetilde{M}_g = 2^k \cdot \sin^2\left(\frac{\theta_g}{2}\right).$$

In order to guarantee the condition $||g|| < 2^k/2$ needed for the quantum counting procedure, we can increase the search space by adding one ancilla qubit (see [12]). We need at least $k + 4m + 1 + w_N + (w_K + 3)$ logical qubits, where in $(w_K + 3)$ we mean 1 ancilla qubit is taken into account for the implementation of the subtraction when implementing g, 1 ancilla qubit to increase the search space and 1 ancilla qubit to implement the Grover iteration relatively g.
4: Apply the Grover algorithm to search for the first round key relative to the Boolean function g. After $\left[\frac{\pi}{4}\sqrt{\frac{2^k}{\widetilde{M}_g}}\right]$ Grover iterations and measurement of qubits with probability $\sin^2\left(\frac{2\left[\frac{\pi}{4}\sqrt{\frac{2^k}{\widetilde{M}_g}}\right]+1}{2}\theta_g\right)$ we get one from $\widetilde{M}_g$ possible solutions, i.e. such a key is K_1' on which $\theta_{K_1'} > \theta_{current}$. Set $\theta_{current} = \theta_{K_1'}$ and repeat the search procedure again. To implement the Grover algorithm relatively g we need $k + 4m + 1 + w_N + 2$ logical qubits, one ancilla qubit needs to implement the subtraction with implementation of g and another one qubit needs to implement the Grover iteration.

If each quantum counting procedure execute correctly, then to search for the key K with the maximum value θ_K, i.e. with the maximum value of $Z(K)$, it is required no more than $\widetilde{M}_g$ iterations of the search procedure (for $\widetilde{M}_g$ obtained at the first start of the search procedure).

The lower bound of search procedure success probability can be estimated by $\frac{4}{\pi^2} \cdot \frac{4}{\pi^2} \cdot 0.5 = 0.0821279$ (this estimate is for the case when the quantum counting procedure performed only twice and the success probability of the Grover's algorithm for searching maximum > 0.5), i.e. on average, it will take about 12 starts to find the new value of $\theta_{current}$.

The complexity of each searching procedure is at least

$$O\left(2^{w_N} + 2^{w_K} + \left[\frac{\pi}{4}\sqrt{\frac{2^k}{\widetilde{M}_g}}\right]\right)$$

quantum operations.

Remark 2 1. The work [17] presents the results of a successful simulation of SDES key searching by using Grover's algorithm in the Quipper quantum simulator. The correct scheme for implementing one Grover iteration is shown in the Fig. 6.

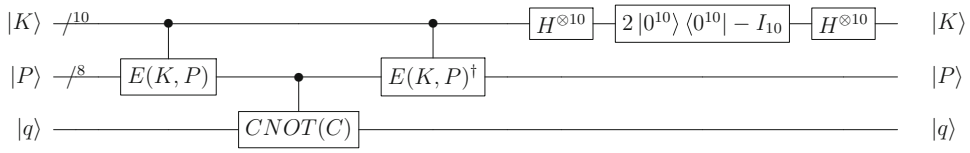


Fig. 6 Implementation of one iteration of the Grover's algorithm for SDES key searching [17]

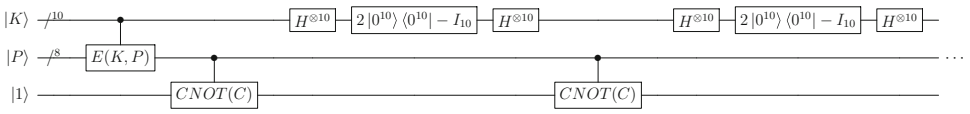


Fig. 7 Wrong iteration of the Grover's algorithm for SDES key searching [17]

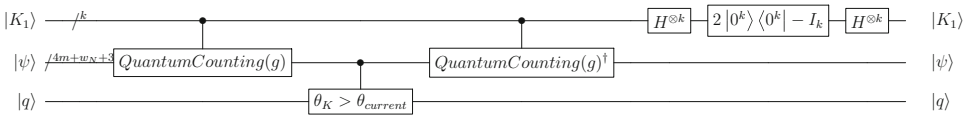


Fig. 8 The correct implementation of one Grover's algorithm iteration with using quantum counting as a "subprogram"

2. It might seems that quantum circuit in the Fig. 7 also work correctly, where the encryption function $E(K, P)$ in Fig. 7 is implemented only once. However, the results of experiments in Quipper quantum simulator indicate that quantum circuit in Fig. 7 does not lead to success, i.e. "inversion about mean" doesn't increase the amplitude of target secret key. Consequently, we have to implement the encryption function $E(K, P)$ as quantum circuit and the inverse quantum circuit $E(K, P)^{\dagger}$ (after inverting the flag qubit) at each iteration of the Grover's algorithm.
   It follows from the fact that the two-qubit CNOT operation swaps the corresponding amplitudes (see [12]):

$$
\begin{aligned}
|q_1 q_2\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle \\
+ \alpha_{11}|11\rangle \xrightarrow{CNOT(q_1|q_2)} \\
\xrightarrow{CNOT(q_1|q_2)} \alpha_{00}|00\rangle + \alpha_{01}|01\rangle \\
+ \alpha_{10}|11\rangle + \alpha_{11}|10\rangle,
\end{aligned}
$$

   therefore, in order for "inversion about mean" increases amplitude of the target secret key, we have to implement $E(K, P)^{\dagger}$.
3. Therefore, from the previous reasoning in remark 2.1 and 2.2, at step 4 of presented search procedure, **quantum counting relatively to the Boolean function f_{K_1} for preparing $\sum_{K_1} \frac{1}{\sqrt{2^k}}|K_1\rangle|\widetilde{\theta}_{K_1}\rangle$ should be performed at each Grover iteration relatively Boolean function g.** The general view of the correct quantum circuit that implements one Grover iteration at step 4 is shown in the Fig. 8.
4. In case when the quantum countings in step 4 are performed at each Grover's iteration, the inversion of the flag qubit in Grover's algorithm that correspond to the indicator of the event $\widetilde{\theta}_{K_1} > \theta_{current}$ is performed with a probability at least $\frac{4}{\pi^2}$.
   Then, if the correct executions of quantum countings are independent events, the lower bound of the success probability of the entire algorithm should be multiplied by $\left(\frac{4}{\pi^2}\right)^{\left[\frac{\pi}{4}\sqrt{\frac{2^k}{M_g}}\right]}$. For simplicity, we will not take this into account, as if the success probability of quantum counting at each Grover's iteration is equal to one.
5. For completeness, we present estimates of complexity in two cases. In the first case, in the author's opinion, the wrong case, the quantum counting at step 4 is performed once, i.e. get an understated estimates of the complexity. In the second case, at step 4, the quantum counting procedure and its inversion is performed at each Grover iteration.

Obtained estimates of the complexity of quantum differential cryptanalysis are more accurate than estimate $O(2^{k/2} + 2^{m/2})$ from [6,7]. If the number of known pairs $N < 2^m$, the scheme for applying the quantum differential cryptanalysis does not change, it is possible to reduce the complexity at the preparation step.

3 Conclusion

1. The complexity of the quantum differential crypanalysis may turn out to be less than the complexity of key searching by the Grover's algorithm, if $E : V_n \times V_m \to V_m$ and $m < n/2$.
2. In case number of pairs $N < 2^m$, the schemes for applying the quantum differential crypanalysis do not change, it is possible to reduce the complexity at the preparatory stage. The required number of logical qubits remains the same as in the case of $N = 2^m$.
3. Acceleration of computations due to "quantum parallelism" in the quantum differential cryptanalysis, when we talk about key search, is apparently absent. Using quantum counting as a "subprogram" of the Grover algorithm eliminates quantum acceleration, since the

Table 3 Estimates of the complexity of the quantum differential cryptanalysis: 1) $O\left(N+2^{w_N}+2^{w_K}+2^{w_K}\cdot 1+\left[\frac{\pi}{4}\sqrt{\frac{2^k}{M_g}}\right]\right)$ - an understated estimates of the complexity, optimistic from the point of view of a quantum cryptanalyst. 2) $O\left(N+2^{w_N}+2^{w_K}+2^{w_K+1}\cdot\left[\frac{\pi}{4}\sqrt{\frac{2^k}{M_g}}\right]\right)$ - the correct estimate. 3) Required at least $k+4m+1+w_N+(w_K+3)$ logical qubits, where $w_N=\lceil m/2\rceil+3$, $w_K=\lceil k/2\rceil+3$

Conditions	The complexity of the quantum differential cryptanalysis (number of quantum operations)
1.At step 4, quantum counting is performed only once (false)	$O\left(N+2^{w_N}+2^{w_K}+2^{w_K}\cdot 1+\left[\frac{\pi}{4}\sqrt{\frac{2^k}{M_g}}\right]\right)$
2.At step 4, quantum counting and its inversion are performed at each Grover iteration (true).	$O\left(N+2^{w_N}+2^{w_K}+2^{w_K+1}\cdot\left[\frac{\pi}{4}\sqrt{\frac{2^k}{M_g}}\right]\right)$

main part of computation complexity described by $O(\sqrt{K})\cdot O(\sqrt{K})\approx O(K)$ quantum operations.

References

1. Biham E., Shamir A.: Differential cryptanalysis of the data encrypt standard. ISBN 978-1-4613-9314-6 (1993)
2. Matsui M.: Linear cryptanalysis method for DES cipher. In: Workshop on the Theory and Application of of Cryptographic Technique, pp. 386–397 (1994)
3. Xie, H., Yang, L.: Using Bernstein-Vazirani algorithm to attack block ciphers. Des. Codes Cryptogr. **87**, 1161–1182 (2019). https://doi.org/10.1007/s10623-018-0510-5
4. Li, H.-W., Yang, L.: A quantum algorithm to approximate the linear structures of Boolean functions. Math. Struct. Comput. Sci. **1**, 1–13 (2014). https://doi.org/10.1017/S0960129516000013
5. Li, H.-W., Yang, L.: Quantum differential cryptanalysis to the block ciphers. Int. Conf. Appl. Tech. Inf. Secur. (2015). https://doi.org/10.1017/978-3-662-48683-2_5
6. Kaplan, M., Leurent, G., Leverrier, A., Naya-Plasencia, M.: Quantum differential and linear cryptanalysis. IACR Trans. Symmetric Cryptol. **1**, 71–94 (2016). https://doi.org/10.13154/tosc.v2016.i1.71-94
7. Zhou, Q., Lu, S., Zhang, A., Sun, J.: Quantum differential cryptanalysis. Quantum Inf. Process. **14**, 2101–2109 (2015). https://doi.org/10.1007/s11128-015-0983-3
8. Grover, L.K.: Quantum mechanics helps in searching for a needle in a haystack. Phys. Rev. Lett. **79**(2), 325 (1997)
9. Brassard, G., Hoyer, P., Tapp, A.: Quantum counting. Automata Lang. Program. **1443**, 820–831 (1998). https://doi.org/10.1007/BFb0055105
10. Denisenko, D.V.: Application of the quantum counting to estimation the weights of Boolean functions in Quipper. J. Exp. Theor. Phys. **130**, 643–648 (2020). https://doi.org/10.1134/S1063776120040032
11. Durr, C., Hoyer, P.: A quantum algorithm for finding the minimum. Phys. Rev. Lett. https://arxiv.org/abs/quant-ph/9607014 (1996)
12. Nielsen, M.A., Chuang, I.L.: Quantum computation and quantum information. Cambridge University Press. http://csis.pace.edu/ctappert/cs837-18spring/QC-textbook.pdf (2010)
13. Denisenko, D.V.: Quantum circuits for S-box implementation without ancilla qubits. J. Exp. Theor. Phys. **128**(6), 847–855 (2019). https://doi.org/10.1134/S1063776119050108
14. Denisenko, D.V., Nikitenkova, M.V.: Optimization of S-boxes GOST R 34.12–2015 "Magma" quantum circuits without ancilla qubits. Matematicheskie Voprosy Kriptografii **11**, 43–52 (2020). https://doi.org/10.4213/mvk312
15. Bernstein E., Vazirani, U.: Quantum complexity theory. In: Proceedings of the 25th Annual ACM Symposium on Theory of Computing, pp. 11–20. https://doi.org/10.1145/167088.167097 (1993)
16. Benenti, G., Casati, G., Strini, G.: Principles of Quantum Computation and Information (2004). https://doi.org/10.1142/5528
17. Denisenko, D.V., Nikitenkova, M.V.: Application of Grover's quantum algorithm for SDES key searching. J. Exp. Theor. Phys. **128**, 25–44 (2019). https://doi.org/10.1134/S1063776118120142
18. Roetteler, M., Steinwandt, R.: A note on quantum related-key attacks. Inf. Process. Lett. **115**, 40–44 (2015). https://doi.org/10.1016/j.ipl.2014.08.009
19. Cuccaro, S.A., Draper, T.G., Kutin, S.A., Moulton, D.P.: A new quantum ripple-carry addition circuit. https://arxiv.org/abs/quant-ph/0410184 (2004)

20. Draper, T.G., Kutin, S.A., Rains, E.M., Svore, K.M.: A logarithmic-depth quantum carry-lookahead adder, Quantum information and computation, vol. 6. https://arxiv.org/abs/quant-ph/0406142 (2004)
21. Kaye, P.: Reversible addition circuit using one ancillary bit with application to quantum computing. https://arxiv.org/abs/quant-ph/0408173 (2004)
22. Brassard, G., Hoyer, P., Mosca, M., Tapp, A.: Quantum amplitude amplification and estimation. AMS Contemporary Mathematics Series, vol. 305, ISBN 9780821821404. https://doi.org/10.1090/conm/305/05215 (2000)

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